MTH203: Sequences and series

Homework 01

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Problem 1. Prove that there is no $x \in \mathbb{Q}$ such that $x^2 = 12$.

Problem 2. Let A and B be countable sets. Prove that $A \cup B$ is countable.

Problem 3. Let $f : A \to B$ and $g : B \to C$ be functions.

(a) Prove that if $g \circ f$ is injective, then f is injective.

(b) Prove that if $g \circ f$ is surjective, then g is surjective.

Problem 4. Use induction to prove that

$$\sum_{k=1}^{n} (2k-1)^2 = 1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}.$$

Problem 5. Let $f : X \to Y$ be a function. Prove that the following two statements are equivalent:

(a) f is injective.

(b) $f(A \cap B) = f(A) \cap f(B)$ for all subsets A, B of X.